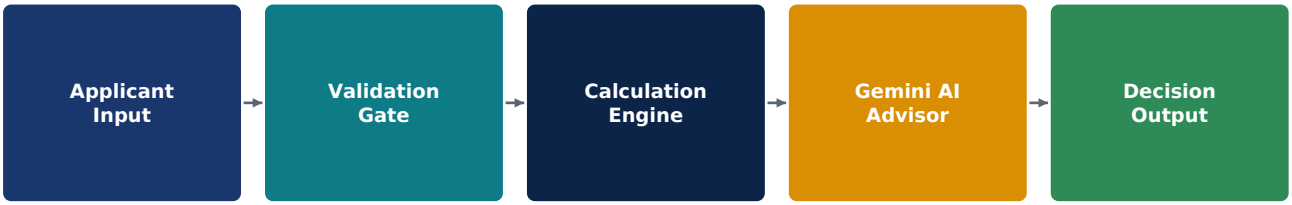
CREDIFY AI

Next-Generation Credit Decisioning and Underwriting Engine

EXPANDED TECHNICAL SPECIFICATION



Hybrid underwriting pipeline: numbers from code, explanations from AI

Item	Detail
Document type	Expanded technical blueprint (extends the 3-page original specification)
Scope	Architecture, mathematics, scenarios, modules, data contracts, roadmap, security and compliance
Market context	India: amounts in INR, FOIR/DTI policy, RBI digital lending rules
Core principle	Deterministic engine calculates; AI layer only explains
Maximum evaluation cap	INR 50,00,000

Reading note. Figures from the original specification are marked **[Spec]**. Figures or policies added during this expansion are marked **[Proposed]** or **[Assumed]** so they can be reviewed and approved separately. Worked numbers were computed with the formulas in Section 3.

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1. Executive Summary and Core Objectives

1.1 Background and Market Context

Credify AI is an enterprise financial intelligence platform that automates credit evaluation, debt capacity modeling and loan eligibility processing [Spec]. Traditional underwriting leans on a single credit-bureau snapshot (a score plus tradeline history). That snapshot has three structural weaknesses:

- **Stale data:** a score reflects past repayment behavior, not current cash flow or today's obligations.
- **Thin-file blindness:** gig workers, new-to-credit applicants and small business owners are scored poorly or not at all, even when they can afford the loan.
- **Inconsistent manual judgment:** two credit officers can reach different decisions on identical files, which is a fairness and audit problem.

Credify AI addresses these by modeling **repayment capacity** (what the applicant can afford now) alongside **repayment history** (what they did before), and by making every decision reproducible.

1.2 Problem Statement

Problem	Legacy impact	Credify AI response
Static bureau snapshots	Misses current income and obligations	Capacity modeled from live income and debt inputs
Non-standard income	Gig and variable earners rejected	Stability Factor adjusts income (scenario SCN-04)
Opaque rejections	Applicants do not know why they failed	AI-generated plain-language risk notes
Manual calculation errors	Wrong EMI, FOIR or eligibility	Deterministic math engine
Slow turnaround	Days of manual review	Sub-second numeric decision
Weak audit trail	Hard to explain a past decision	Versioned rules, replayable inputs and outputs

1.3 The Hybrid Underwriting Principle

The central design rule is a hard separation of responsibilities. The **Deterministic Engine** computes every number (FOIR, DTI, EMI, maximum principal) and the decision status. The **AI Advisory Layer** (Gemini) only explains values it is handed. It never calculates, never changes a status and never invents a figure.

Figure 1.1: Separation of authority

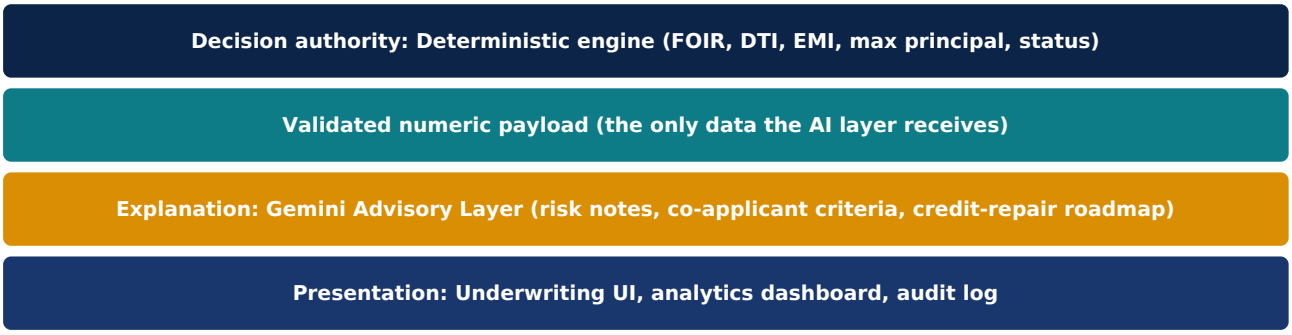


Figure 1.1: Numbers flow downward; the AI layer cannot write back to the decision

The practical consequence is audit-friendliness. Any decision can be replayed from the stored inputs and rule version and will give the identical numeric result. An LLM-only decision cannot guarantee that.

1.4 Headline Metrics Explained

Metric [Spec]	Meaning	Engineering note
Max evaluation cap INR 50,00,000	Ceiling on any evaluated principal	Applied as min(formula principal, cap)
Engine latency under 850 ms	End-to-end response budget	See Section 1.5 for the budget split
Calculation 100% deterministic	Same input gives same output	Pure functions, no randomness, no clock reads
AI: Gemini Flash tier	Low-latency explanation model	Model name kept in configuration, not hard-coded

Worked example of the cap. Take scenario SCN-01 (available EMI INR 57,500) at an assumed 10.5% per annum over 240 months [Assumed]:

```
def max_principal(emi, annual_rate, months):
    r = annual_rate / 12 / 100
    return emi * (1 - (1 + r) ** -months) / r

raw    = max_principal(57_500, 10.5, 240)
capped = min(raw, 50_00_000)
print(f"Raw formula principal : {raw:,.0f}")
print(f"After INR 50L cap     : {capped:,.0f}")
```

```
# Output
Raw formula principal : 5,759,331
After INR 50L cap     : 5,000,000
```

The applicant's capacity (about INR 57.6 lakh) exceeds the product ceiling, so the cap binds. The EMI for INR 50,00,000 at these terms is about INR 49,919, comfortably inside the INR 57,500 headroom.

1.5 Latency Budget

The specification states the calculation engine runs in under 10 ms and the whole engine in under 850 ms. The unstated risk is that a large-language-model API call can take longer than the rest of the pipeline combined. A realistic budget is:

Stage	Target	Notes
Validation gate	under 20 ms	Schema parse plus sanity rules
Calculation engine	under 10 ms [Spec]	Closed-form annuity math
Decision and status	under 5 ms	Rule lookup, no I/O
Persist audit record	under 30 ms	Append-only write, may be async
Gemini advisory call	variable	Often the dominant cost; run asynchronously

> **Recommendation [Proposed].** Return the numeric decision synchronously and attach the AI notes asynchronously (or stream them). This keeps the sub-second claim true for the decision itself, which is what downstream systems depend on.

1.6 Objectives

- **Accuracy:** zero arithmetic error in credit mathematics.
- **Speed:** instant numeric decisions for straightforward files.
- **Explainability:** every decision carries a plain-language rationale.
- **Inclusion:** a defined, rule-based path for variable-income applicants.
- **Auditability:** inputs, rule versions and outputs logged for exact replay.
- **Compliance:** alignment with Indian digital-lending rules (Section 7).

Non-goals. The engine does not replace bureau pulls, fraud detection, KYC, collateral valuation or legal documentation. It consumes the outputs of those systems as inputs.

1.7 Deep Research: Applicant (Candidate) Profiles

The four scenarios in the specification map to four applicant archetypes. Each carries a different risk and needs different treatment.

Archetype	Income pattern	Main risk	Engine treatment
Salaried (SCN-01, SCN-03)	Fixed, verifiable via payslips	Over-leverage	Standard FOIR cap (55% in SCN-01)
Self-employed / retailer (SCN-02)	Lumpy, partly cash	Understated obligations	Stricter FOIR cap (50%)
Gig / consultant (SCN-04)	Variable month to month	Income volatility	Stability Factor (0.75)
Thin-file or young applicant	Short history	Little data to score	More weight on capacity, less on score

FOIR caps differ by archetype (55% versus 50%), showing that policy is a **lookup table**, not one constant. This is a sound design and should be explicit in configuration (Section 3.3).

Applicant data dimensions

Dimension	Examples	Used for
Income	Monthly gross, employment type, variability	FOIR cap, capacity, stability adjustment
Obligations	Existing EMIs, card minimums, rent where policy requires	Available EMI, DTI
Request	Amount, tenure, interest rate	Requested EMI, affordability test
Credit quality	Bureau score estimate, delinquencies	Eligibility floor, pricing, advisory notes
Context	Co-applicant, collateral, existing relationship	Mitigation options

1.8 Stakeholders

Stakeholder	Interest
Credit officers	Fast, consistent, explainable decisions they can defend
Risk and compliance	Audit trail, policy versioning, change control
Applicants	Clear reasons and a concrete path to improve
Engineering	Testable, modular, loosely coupled components
Regulators (RBI)	Fair practices, disclosure, data protection, grievance redress

1.9 Issues in the Source Specification

A review of the original document found the following. These are flagged so the final version can be corrected.

- **Arithmetic checks out:** SCN-01 (82,500 minus 25,000 equals 57,500) and SCN-02 (37,500 minus 42,000 equals minus 4,500) are correct.
- **SCN-04 omits its FOIR:** the INR 18,750 result only works with a 50% FOIR: 90,000 x 0.75 = 67,500; x 50% = 33,750; minus 15,000 = 18,750. The specification should state this.
- **SCN-03 inputs are incomplete:** a 41.2% DTI with a 27,800 EMI implies about INR 17,520 of existing obligations, which is not shown. Rate and tenure behind the EMI are also missing (about 9.4% over 20 years reproduces it).
- **Overclaims:** 'zero hallucination' and '100% transparent' are too strong. Constraining AI inputs reduces risk but does not eliminate it. Safer wording: numerically grounded, with output validation that every figure in AI text matches engine output.

- **Model naming:** the AI model name should be verified against the provider's current list and held in configuration.

2. High-Level Pipeline Architecture

The execution pipeline separates input ingestion, mathematical calculation, risk scoring and generative explanation into isolated stages [Spec]. Isolation means a failure or change in one stage (for example, swapping the AI model) cannot corrupt the numbers produced by another.

Figure 2.1: Five-stage pipeline



Figure 2.1: Pipeline stages. Only validated numbers reach the AI stage

2.1 Stage 1: Applicant Input

Income and debt data arrive from an application form, a loan origination system or an API client. The stage accepts raw values and performs no calculation. Its only job is to capture the payload and attach metadata (applicant ID, request ID, timestamp, source channel).

- **Inputs:** monthly gross income, existing monthly obligations, requested amount, tenure, interest rate, credit score estimate.
- **Outputs:** a raw payload and a request envelope.
- **Failure modes:** missing fields, wrong types, amounts typed in lakh instead of rupees, duplicate submissions.
- **Controls:** idempotency key per request, client-side hints, server-side re-validation.

2.2 Stage 2: Validation Gate (Schema and Sanity Check)

The gate rejects bad data before any math runs. It has two layers: **schema validation** (types, required fields, ranges) and **sanity validation** (cross-field logic and plausibility).

Check	Type	Rule [Proposed]	On failure
monthlyGrossIncome	Schema	number, greater than 0, finite	Reject: INVALID_INCOME
existingMonthlyObligations	Schema	number, zero or more	Reject: INVALID_OBLIGATIONS
requestedTenureMonths	Schema	integer, 6 to 360	Reject: INVALID_TENURE
interestRatePerAnnum	Schema	number, greater than 0, up to 60	Reject: INVALID_RATE
creditScoreEstimate	Schema	integer, 300 to 900	Reject: INVALID_SCORE
Obligations vs income	Sanity	obligations not above 3 x income	Flag for manual review
Requested amount vs income	Sanity	amount not above 120 x monthly income	Flag: implausible request
Rate vs market band	Sanity	within configured band for product	Flag: pricing anomaly

```
function validate(p: CredifyEvaluationPayload): ValidationResult {
  const errors: string[] = [];
  if (!(p.monthlyGrossIncome > 0)) errors.push("INVALID_INCOME");
  if (p.existingMonthlyObligations < 0) errors.push("INVALID_OBLIGATIONS");
  if (!Number.isInteger(p.requestedTenureMonths) ||
      p.requestedTenureMonths < 6 || p.requestedTenureMonths > 360)
    errors.push("INVALID_TENURE");
  if (!(p.interestRatePerAnnum > 0 && p.interestRatePerAnnum <= 60))
    errors.push("INVALID_RATE");
  return { ok: errors.length === 0, errors };
}
```

2.3 Stage 3: Financial Calculation Engine

The engine evaluates FOIR, DTI, maximum EMI, maximum loan principal and the EMI for the requested loan. It uses only the validated payload and a versioned policy table. It must be a **pure function**: no network calls, no random numbers, no reads of the system clock. That property is what makes replay and regression testing exact.

- **Inputs:** validated payload, policy table (FOIR caps, stability factors, thresholds), rule version.
- **Outputs:** foirPercentage, maxEligibleEMI, maxEligibleLoanAmount, calculatedRequestedEMI, status, reason codes.
- **Latency:** under 10 ms [Spec].
- **Numerics:** use decimal arithmetic or integer paise for money; round only at defined output points (Section 3.7).

2.4 Stage 4: Gemini AI Underwriter (Risk Notes and Strategy)

The AI stage receives a structured prompt built from engine outputs. It returns plain-language risk notes, co-applicant criteria and a credit-repair roadmap. Three safeguards keep it bounded:

- **Input restriction:** the prompt contains only validated numbers and reason codes, never free-form applicant text that could carry prompt injection.
- **Output restriction:** a structured response schema (JSON) with length limits, no decision field.
- **Output verification:** every number in the generated text is extracted and compared with the engine output; a mismatch discards the AI text and falls back to a template.

```
SYSTEM: You explain credit decisions. Use ONLY the figures provided.
Never compute, estimate, or change any number. Never state a decision
different from the one provided.

DATA (verified): status=CONDITIONAL, foir=41.2, maxEmi=42980,
requestedEmi=27800, reasonCodes=[DTI_ABOVE_COMFORT_BAND]
TASK: Return JSON {summary, riskNotes[], coApplicantCriteria[],
improvementSteps[]} in plain language, max 120 words per field.
```

2.5 Stage 5: Underwriting Output (Sanction or Mitigation)

The final stage assembles the decision packet: numeric results, status, reason codes, AI notes and an audit reference. Depending on status the output is a sanction offer, a conditional offer with mitigation steps (lower amount, longer tenure, co-applicant) or a decline with improvement guidance.

2.6 Data Flow and Sequence

Step	From	To	Content
1	Client	API gateway	CredifyEvaluationPayload
2	Gateway	Validation gate	Raw payload plus request envelope

Step	From	To	Content
3	Validation gate	Calc engine	Validated payload (or error response)
4	Calc engine	Audit log	Inputs, rule version, outputs, hash
5	Calc engine	Client	Numeric decision (synchronous)
6	Calc engine	Gemini layer	Verified numbers and reason codes
7	Gemini layer	Verifier	Draft advisory text
8	Verifier	Client / log	Approved advisory text or template fallback

2.7 Failure Handling and Degradation

Failure	Behavior	Customer impact
Validation error	Return structured error; no engine call	Clear message on what to fix
Engine exception	Fail closed: no automated decision; route to manual review	Delay, never a wrong approval
Gemini timeout or error	Use deterministic template notes	Less personalized text, same decision
AI numeric mismatch	Discard AI text; log incident; template fallback	None visible
Audit write failure	Retry queue; block sanction release until persisted	Short delay
Policy table unavailable	Use last known good signed version; alert	None

> **Principle.** The system fails **closed** on anything affecting money (no approval without a verified calculation and a persisted audit record) and fails **open** on cosmetics (AI prose can fall back to a template).

2.8 Design Rationale: Why Isolated Stages

- **Testability:** each stage has its own contract and can be tested alone.
- **Replaceability:** the AI model can be changed without touching decision logic.
- **Regulatory clarity:** the decision authority is a documented, deterministic component.
- **Scaling:** the engine is stateless and horizontally scalable; the AI stage can scale or queue independently.

3. Mathematical Risk Models

The Central Calculation Engine enforces exact credit equations to establish total capacity, debt ratios and maximum principal eligibility [Spec]. This section explains each formula, derives the annuity relationship, defines the policy table and then works every scenario.

3.1 Core Underwriting Formulas

Formula [Spec]	Purpose
Max Allowed EMI = (Monthly Gross Income x FOIR Cap) minus Existing Monthly Obligations	Monthly installment room left after existing debt
Max Loan Principal = Max EMI x (1 - (1 + r)^-n) / r	Largest principal that Max EMI can repay over n months at monthly rate r
DTI (%) = (Total Monthly Obligations / Monthly Gross Income) x 100	Share of income consumed by debt service

Symbols: **r** is the monthly interest rate (annual rate divided by 12 and by 100); **n** is the tenure in months; **EMI** is the equated monthly installment; **P** is the principal.

3.2 Deriving the Annuity Formula

A loan of principal P is repaid by n equal installments E. Each installment is discounted back to today at the monthly rate r. The principal equals the present value of all installments:

```
P = E/(1+r) + E/(1+r)^2 + ... + E/(1+r)^n

This is a geometric series with first term E/(1+r) and ratio 1/(1+r):

P = E * [ 1 - (1+r)^-n ] / r          (maximum principal for a given EMI)

Solving for E:

E = P * r / [ 1 - (1+r)^-n ]          (EMI for a given principal)
```

Both directions are needed. The engine uses the first form to find the biggest loan an applicant can carry (eligibility) and the second to price the loan the applicant actually requested (affordability).

Edge case: zero interest

When the rate is zero the formula divides by zero. The limit as r tends to 0 is P = E x n, and E = P / n. The engine must branch explicitly. Although the validation gate rejects zero rates for standard products, a subsidized scheme could legitimately use one.

```
def emi(P, annual_rate, n):
    if annual_rate == 0:
        return P / n
    r = annual_rate / 1200
    return P * r / (1 - (1 + r) ** -n)

print(round(emi(1_000_000, 10, 12)))     # 1 lakh x 10 for 12 months
print(round(emi(1_200_000, 0, 24)))      # zero-rate branch
```

```
# Output
87916
50000
```

3.3 FOIR: Policy Design

FOIR (Fixed Obligation to Income Ratio) caps the share of income that all fixed debt payments, including the proposed EMI, may consume. The specification uses 55% for a salaried applicant and 50% for a retailer or owner. Lenders typically tier FOIR by income level and employment stability. A configurable table is therefore recommended.

Applicant category	FOIR cap	Basis
Salaried, stable employer (SCN-01)	55% [Spec]	Verifiable fixed income
Self-employed / retailer (SCN-02)	50% [Spec]	Less predictable cash flow
Gig / consultant (SCN-04)	50% [Assumed]	Needed to reproduce INR 18,750
Salaried, high income band	60% [Proposed]	Larger absolute residual income
Thin-file / new-to-credit	45% [Proposed]	Conservative where data is scarce

Governance. FOIR tables are policy, not code. Store them as versioned, signed configuration with an effective date, approver and change log. Every decision record stores the version used.

3.4 DTI: Meaning and Bands

DTI measures total monthly obligations (existing plus proposed) against gross income. FOIR is a hard limit; DTI is also used as a comfort measure inside the limit. The original specification reports DTI for SCN-03 but does not state the band that triggers CONDITIONAL. The following bands are proposed:

Post-loan DTI	Band	Effect [Proposed]
up to 35%	Comfortable	APPROVED if other tests pass
above 35% to 40%	Moderate	APPROVED with advisory note
above 40% up to FOIR cap	Stretched	CONDITIONAL (mitigation required)
above FOIR cap	Breach	DECLINED or reduced-amount counter-offer

3.5 Stability Factor for Variable Income

For gig and consulting income, the engine multiplies reported income by a Stability Factor between 0 and 1 before applying FOIR. The specification uses 0.75 for SCN-04. A proposed way to derive the factor from history:

```
stability = clamp( 1 - coefficient_of_variation(last_12_months_income), 0.50, 1.00 )

# CV = standard deviation / mean. Lower variability gives a factor nearer 1.
# Example: monthly CV of 0.25 gives stability = 0.75
```

Income variability (CV)	Stability factor [Proposed]	Interpretation
up to 0.10	0.90 to 1.00	Near-fixed income
0.10 to 0.25	0.75 to 0.90	Moderate variation
0.25 to 0.50	0.50 to 0.75	High variation
above 0.50	0.50 (floor)	Manual review recommended

3.6 Operational Scenario Assessment (Worked in Full)

The specification lists four scenarios. The table reproduces them; the subsections below show every step and supply the inputs the original left implicit (marked [Assumed]).

ID	Applicant	Financial input [Spec]	Engine output [Spec]	Decision
SCN-01	Salaried IT Lead	Income 1,50,000; debt 25,000	FOIR cap 55% = 82,500; available EMI 57,500	APPROVED
SCN-02	Retailer / Owner	Income 75,000; debt 42,000	FOIR cap 50% = 37,500; deficit -4,500	DECLINED
SCN-03	Product Manager	Income 1,10,000; request 30,00,000	EMI 27,800; DTI 41.2%	CONDITIONAL
SCN-04	Gig Consultant	Variable income 90,000; debt 15,000; stability 0.75	Adjusted capacity 18,750	CAP ADJUSTED

Figure 3.1: Available EMI capacity per scenario (INR per month)

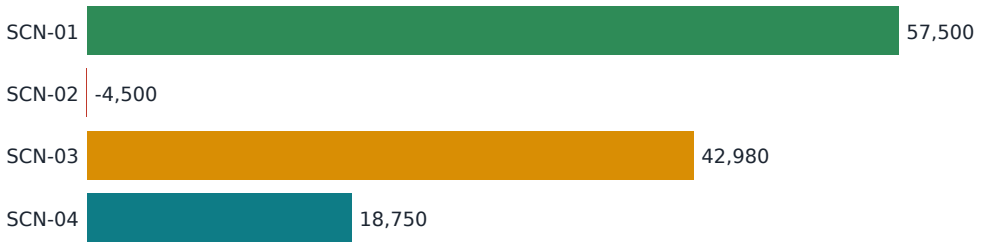


Figure 3.1: SCN-02 shows no bar because its capacity is negative (-4,500); SCN-03 capacity is derived in 3.6.3

3.6.1 SCN-01: Salaried IT Lead (APPROVED)

```
FOIR cap   = 1,50,000 x 55%        = 82,500
Available  = 82,500 - 25,000       = 57,500
Existing DTI = 25,000 / 1,50,000   = 16.7%
Max principal @10.5%, 240 months   = 57,59,331  -> capped to 50,00,000
Illustrative request 30,00,000: EMI = 29,952 (rounded up)
Post-loan DTI = (25,000 + 29,952) / 1,50,000 = 36.6%   (moderate band)
Status: APPROVED, with note that the product cap binds before capacity does
```

Insight: the applicant is capacity-rich. The advisory layer should say so and suggest the applicant could consider a second product or a larger tenure-linked offer if the cap is later raised.

3.6.2 SCN-02: Retailer / Owner (DECLINED)

```
FOIR cap   = 75,000 x 50%          = 37,500
Available  = 37,500 - 42,000       = -4,500   (capacity deficit)
Existing DTI = 42,000 / 75,000     = 56.0%    (already above the FOIR cap)
Max principal                      = 0
Status: DECLINED  (reason NO_CAPACITY)
```

Path to approval [Proposed]. Suppose the applicant wants INR 5,00,000 at 12% over 60 months (EMI about 11,122). Total obligations must then be at most 37,500, so existing debt must fall to about 26,378: a reduction of roughly INR 15,622 per month, for example by closing or consolidating one loan. The advisory layer should present this concrete target rather than a generic refusal.

3.6.3 SCN-03: Product Manager (CONDITIONAL)

```
Inputs [Assumed]: FOIR 55%, rate 9.4%, tenure 240 months
Existing obligations (derived): 41.2% x 1,10,000 - 27,800 = about 17,520
FOIR cap   = 1,10,000 x 55%        = 60,500
Available  = 60,500 - 17,520       = 42,980
Requested EMI (30,00,000)          = 27,769  (spec shows 27,800)
FOIR test: 27,769 <= 42,980        -> pass
DTI = (17,520 + 27,769) / 1,10,000 = 41.2%  -> above 40% comfort band
Status: CONDITIONAL
```

Mitigation options [Proposed]. (a) Extend the tenure to 300 months: EMI falls to 26,003 and DTI to 39.6%, which moves the status to APPROVED. (b) Reduce the amount to about INR 28.6 lakh at the original tenure. (c) Add a co-applicant whose income lifts the combined denominator. The engine should compute and offer option (a) and (b) automatically.

3.6.4 SCN-04: Gig Consultant (CAP ADJUSTED)

```
Adjusted income = 90,000 x 0.75    = 67,500
FOIR cap   = 67,500 x 50%          = 33,750   [FOIR 50% assumed]
Available  = 33,750 - 15,000       = 18,750   (matches spec)
Max principal @12%:  60 months -> 8,42,906
                     84 months -> 10,62,158
                    120 months -> 13,06,885
DTI on adjusted income = 15,000 / 67,500 = 22.2%  (16.7% on gross)
Status: CAP ADJUSTED
```

Insight: reporting both DTI values (adjusted and gross) avoids confusion. Policy should state which one governs decisions; this document proposes the adjusted figure.

3.7 Rounding and Numeric Precision

Quantity	Rule [Proposed]	Reason
Requested EMI	Round up to next rupee	Never understate the installment
Max EMI	Round down	Never overstate capacity
Max principal	Round down to nearest rupee, then apply cap	Conservative eligibility
Ratios (FOIR, DTI)	Compute unrounded; display one decimal	Avoid threshold errors from rounding
Storage	Integer paise or decimal type, never binary floats for money	Exact replay

Why it matters. A DTI of 40.04% displayed as 40.0% must not be classified as 'within 40%'. Compare the unrounded value to the threshold, and show the rounded value only to humans.

3.8 Sensitivity Analysis (SCN-01 capacity, INR 57,500 EMI)

Interest rate (240 months)	Max principal	Tenure (10.5%)	Max principal
9.5%	61,68,660	120 months	42,61,311
10.5%	57,59,331	180 months	52,01,742
11.5%	53,91,823	240 months	57,59,331
12.5%	50,60,994	300 months	60,89,929

Figure 3.2: Max principal vs interest rate (INR lakh, 240 months)



Figure 3.2: Each +1 point of rate removes roughly INR 3.3 to 4.1 lakh of capacity

Two conclusions follow. First, a one-point rate change moves eligibility by roughly 6%, so rate data must be accurate and timestamped. Second, longer tenure increases principal with diminishing returns (180 to 240 months adds 5.6 lakh; 240 to 300 adds 3.3 lakh) while total interest paid rises sharply, which the advisory layer should mention.

Cost of tenure extension (INR 50 lakh at 10.5%)

Quantity	Value
EMI over 240 months	about 49,919
Total paid over 240 months	about 1,19,80,559
Total interest over 240 months	about 69,80,559
Month-1 split	interest 43,750; principal 6,169
Month-2 split	interest 43,696; principal 6,223

Early installments are mostly interest. This is a fairness-in-disclosure point: the Key Facts Statement shown to the borrower (Section 7) must make total cost visible.

3.9 Decision Logic and Reason Codes

```
IF availableEMI <= 0                      -> DECLINED  (NO_CAPACITY)
ELSE IF requestedEMI > availableEMI       -> DECLINED  (EMI_EXCEEDS_CAPACITY)
ELSE IF score < minimumScore              -> DECLINED  (SCORE_BELOW_FLOOR)
ELSE IF postLoanDTI > comfortBand (40%)   -> CONDITIONAL (DTI_ABOVE_COMFORT_BAND)
ELSE IF stabilityFactor < 1               -> CAP_ADJUSTED (INCOME_VARIABILITY)
ELSE                                      -> APPROVED
```

Reason code	Meaning	Typical mitigation
NO_CAPACITY	Existing debt already exceeds FOIR cap	Reduce obligations, add co-applicant
EMI_EXCEEDS_CAPACITY	Requested EMI above available EMI	Lower amount or longer tenure
SCORE_BELOW_FLOOR	Credit score below policy minimum	Credit-repair roadmap, secured product
DTI_ABOVE_COMFORT_BAND	Within FOIR but stretched	Extend tenure, reduce amount, co-applicant
INCOME_VARIABILITY	Income adjusted by stability factor	Provide longer income history
PRODUCT_CAP_APPLIED	Evaluation cap bound the amount	Alternate product if available

3.10 Reference Implementation (Python)

```
import math
FOIR = {"SALARIED": 0.55, "SELF_EMPLOYED": 0.50, "GIG": 0.50}
CAP = 5_000_000

def emi(P, rate, n):
    if rate == 0: return P / n
    r = rate / 1200
    return P * r / (1 - (1 + r) ** -n)

def max_principal(E, rate, n):
    if rate == 0: return E * n
    r = rate / 1200
    return E * (1 - (1 + r) ** -n) / r

def evaluate(income, debt, cat, rate, n, req=None, stability=1.0):
    adj = income * stability
    cap = adj * FOIR[cat]
    avail = cap - debt
    out = {"foir_cap": round(cap), "avail_emi": round(avail)}
    if avail <= 0:
        out.update(status="DECLINED", reason="NO_CAPACITY", max_loan=0)
        return out
    out["max_loan"] = math.floor(min(max_principal(avail, rate, n), CAP))
    if req:
        e = math.ceil(emi(req, rate, n))
        dti = (debt + e) / income * 100
        out.update(req_emi=e, dti=round(dti, 1))
        out["status"] = ("DECLINED" if e > avail else
                         "CONDITIONAL" if dti > 40 else "APPROVED")
    else:
        out["status"] = "CAP_ADJUSTED" if stability < 1 else "APPROVED"
    return out
```

Labeled test run (all five cases executed)

```
SCN-01  evaluate(150000, 25000, "SALARIED", 10.5, 240, req=3_000_000)
 -> foir_cap 82500, avail_emi 57500, max_loan 5000000, req_emi 29952, dti 36.6, APPROVED
SCN-02  evaluate(75000, 42000, "SELF_EMPLOYED", 12, 60, req=500_000)
 -> foir_cap 37500, avail_emi -4500, DECLINED (NO_CAPACITY)
SCN-03  evaluate(110000, 17520, "SALARIED", 9.4, 240, req=3_000_000)
 -> avail_emi 42980, max_loan 4643420, req_emi 27769, dti 41.2, CONDITIONAL
SCN-03b evaluate(110000, 17520, "SALARIED", 9.4, 300, req=3_000_000)
 -> req_emi 26003, dti 39.6, APPROVED   (tenure-extension mitigation)
SCN-04  evaluate(90000, 15000, "GIG", 12, 60, stability=0.75)
 -> foir_cap 33750, avail_emi 18750, max_loan 842906, CAP_ADJUSTED
```

3.11 Summary Comparison of the Four Scenarios

Scenario	Income basis	FOIR	Available EMI	Post-loan DTI	Status
SCN-01	1,50,000	55%	57,500	36.6% (30L request)	APPROVED
SCN-02	75,000	50%	-4,500	56.0% (existing)	DECLINED
SCN-03	1,10,000	55%	42,980	41.2%	CONDITIONAL
SCN-04	67,500 adjusted	50%	18,750	22.2% (adjusted, existing)	CAP ADJUSTED

4. Core System Modules

The platform is organized into four modules [Spec]. This section expands each into responsibilities, interfaces, internal components and quality attributes.

Figure 4.1: Module stack

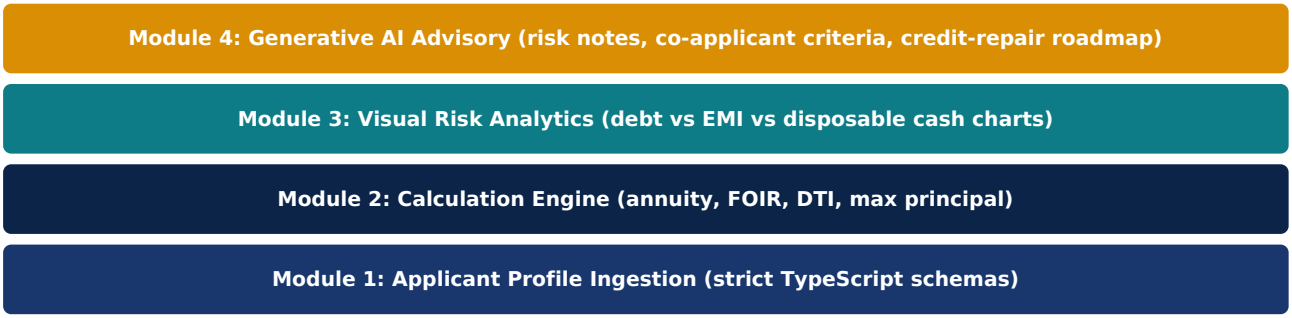


Figure 4.1: Modules build upward; data flows from ingestion to advisory

4.1 Module 1: Applicant Profile Ingestion

Captures monthly gross income, obligations, credit bureau scores and loan requests using strict TypeScript schemas [Spec].

Aspect	Detail
Responsibility	Receive, type-check, normalize and stamp the applicant payload
Inputs	Web form, partner API, batch file import, loan origination system
Outputs	CredifyEvaluationPayload plus request envelope
Key components	Schema validators, unit normalizer (lakh/crore to rupees), duplicate detector, PII tagger
Quality attributes	Strictness (reject early), idempotency, privacy by design
Risks	Unit confusion, partial submissions, injection through free-text fields

- **Normalization:** reject ambiguous formats rather than guess; show the parsed value back to the user.
- **Bureau score handling:** store the score and bureau date; scores older than a policy window trigger a refresh.
- **PII tagging:** mark identifying fields at ingestion so masking and retention rules apply downstream.

4.2 Module 2: Calculation Engine

Executes deterministic annuity formulas, debt service limits and maximum principal caps in under 10 milliseconds [Spec].

Aspect	Detail
Responsibility	All numerical results and the decision status
Inputs	Validated payload; versioned policy table
Outputs	UnderwritingOutput numeric fields, status, reason codes, rule version
Key components	Annuity functions, FOIR/DTI calculators, stability adjuster, decision rules, rounding policy
Quality attributes	Purity, determinism, speed, exhaustive test coverage
Risks	Float rounding, silent policy changes, divide-by-zero at zero rate

- **Property tests:** repaying a computed EMI over n months must reduce the balance to zero within one paisa; max_principal(emi(P)) must return P.
- **Monotonicity tests:** higher income never lowers capacity; higher rate never raises max principal.

- **Golden files:** the four scenarios plus the 500+ synthetic profiles of Milestone 5 are stored with expected outputs.

4.3 Module 3: Visual Risk Analytics

Renders real-time visual comparisons between existing debt, proposed installments and net disposable cash reserves [Spec].

Visual	Shows	Audience
FOIR gauge	Used versus allowed share of income	Credit officer, applicant
Debt composition bar	Existing obligations versus proposed EMI versus residual income	Credit officer
Scenario slider	How amount, tenure and rate change EMI and status	Applicant (what-if)
Status badge	APPROVED / CONDITIONAL / DECLINED with reason codes	Everyone
Sensitivity chart	Max principal against rate and tenure	Risk analyst

Design rule [Proposed]. Charts must read directly from engine output and never recompute in the browser. Recomputation in two places is a classic source of numbers that disagree between screen and record.

4.4 Module 4: Generative AI Advisory

Interprets verified calculation outputs to produce plain-language risk notes, co-applicant criteria and credit-repair roadmaps [Spec].

Aspect	Detail
Responsibility	Explain the decision in clear language and suggest next steps
Inputs	Verified numbers, status, reason codes, applicant category
Outputs	Structured JSON notes (summary, risk notes, co-applicant criteria, improvement steps)
Key components	Prompt templates, schema-constrained generation, numeric verifier, template fallback, safety filter
Quality attributes	Faithfulness, readability, tone, language support, bounded latency
Risks	Invented figures, discriminatory phrasing, prompt injection, promising approval

- **No-promise rule:** advisory text must not state or imply an approval beyond the engine's status.
- **Fairness rule:** notes may refer to financial factors only; no references to gender, religion, caste, location proxies or similar attributes.
- **Language:** support English and major Indian languages through the same verified numbers with translated templates.
- **Evaluation:** sample outputs weekly; measure numeric-match rate (target 100%), readability and officer-rated usefulness.

5. Application Data Schema Contract

The contract defines exactly what enters and leaves the engine. The original specification gives two TypeScript interfaces [Spec]; they are reproduced here in readable form and then extended with field-level rules, validation code, API shapes and an audit record.

5.1 Core Interfaces [Spec]

```
interface CredifyEvaluationPayload {
  applicantId: string;
  monthlyGrossIncome: number;
  existingMonthlyObligations: number;
  requestedLoanAmount: number;
  requestedTenureMonths: number;
  interestRatePerAnnum: number;
  creditScoreEstimate: number;
}

interface UnderwritingOutput {
  foirPercentage: number;
  maxEligibleEMI: number;
  maxEligibleLoanAmount: number;
  calculatedRequestedEMI: number;
  status: 'APPROVED' | 'CONDITIONAL' | 'DECLINED';
  aiAdvisorNotes: string;
}
```

5.2 Field Dictionary

Field	Type	Unit	Rule [Proposed]	Used by
applicantId	string	-	opaque ID, no PII inside	Audit, masking
monthlyGrossIncome	number	INR / month	greater than 0	FOIR, DTI
existingMonthlyObligations	number	INR / month	0 or more	Available EMI, DTI
requestedLoanAmount	number	INR	0 to 50,00,000 evaluated	Requested EMI
requestedTenureMonths	number	months	integer 6 to 360	Annuity
interestRatePerAnnum	number	percent	greater than 0 to 60	Annuity
creditScoreEstimate	number	points	integer 300 to 900	Floor check, notes
foirPercentage	number	percent	0 to 100+	Display, audit
maxEligibleEMI	number	INR / month	rounded down	Display
maxEligibleLoanAmount	number	INR	rounded down, capped	Display
calculatedRequestedEMI	number	INR / month	rounded up	Display, KFS
status	enum	-	three values	Workflow
aiAdvisorNotes	string	-	verified text or template	Display

5.3 Recommended Contract Extensions

The original interfaces lack several fields needed for a production system. Each addition below is a proposal.

Addition	Reason
employmentType: 'SALARIED' | 'SELF_EMPLOYED' | 'GIG'	Selects FOIR cap and stability logic (SCN-01 to SCN-04)
incomeHistory?: number[] (last 12 months)	Computes the stability factor for variable income
coApplicant?: { income, obligations }	Supports mitigation for CONDITIONAL cases
reasonCodes: string[] in output	Machine-readable explanation; feeds the AI layer
policyVersion: string in output	Exact replay and audit
requestId and idempotencyKey	Duplicate protection, tracing
currency and country	Future multi-market support
status 'CAP_ADJUSTED'	Spec scenario SCN-04 uses it but the interface lacks it

Contract gap. Scenario SCN-04 reports CAP ADJUSTED, but the status union in the interface has only three values. Either add a fourth status or represent it as APPROVED or CONDITIONAL with the reason code INCOME_VARIABILITY. This should be decided and documented.

5.4 Validation Implementation

```
import { z } from "zod";

export const PayloadSchema = z.object({
  applicantId: z.string().min(1).max(64),
  employmentType: z.enum(["SALARIED", "SELF_EMPLOYED", "GIG"]),
  monthlyGrossIncome: z.number().positive().finite(),
  existingMonthlyObligations: z.number().nonnegative().finite(),
  requestedLoanAmount: z.number().positive().max(5_000_000),
  requestedTenureMonths: z.number().int().min(6).max(360),
  interestRatePerAnnum: z.number().positive().max(60),
  creditScoreEstimate: z.number().int().min(300).max(900),
}).strict();   // unknown fields are rejected, not ignored
```

5.5 API Contract

Endpoint [Proposed]	Method	Purpose
/v1/evaluations	POST	Submit payload; returns numeric decision synchronously
/v1/evaluations/{id}	GET	Fetch decision and (when ready) advisory notes
/v1/evaluations/{id}/replay	POST	Re-run with stored inputs and rule version for audit
/v1/policies/current	GET	Read active policy version metadata
/v1/health	GET	Liveness and dependency status

Example request and response (SCN-03)

```
POST /v1/evaluations
{ "applicantId": "A-10293", "employmentType": "SALARIED",
  "monthlyGrossIncome": 110000, "existingMonthlyObligations": 17520,
  "requestedLoanAmount": 3000000, "requestedTenureMonths": 240,
  "interestRatePerAnnum": 9.4, "creditScoreEstimate": 752 }

200 OK
{ "requestId": "r-77f1", "policyVersion": "2026.10.1",
  "foirPercentage": 41.2, "maxEligibleEMI": 42980,
  "maxEligibleLoanAmount": 4643420, "calculatedRequestedEMI": 27769,
  "status": "CONDITIONAL", "reasonCodes": ["DTI_ABOVE_COMFORT_BAND"],
  "aiAdvisorNotes": null, "aiStatus": "PENDING" }
```

Note that the engine response returns **before** the AI notes exist. The client polls or subscribes for the advisory text, consistent with the latency design in Section 1.5. In the example, the field named foirPercentage carries the post-loan obligations ratio (41.2%); the contract should define this meaning explicitly to avoid confusion with DTI.

Error response

```
422 Unprocessable Entity
{ "error": "VALIDATION_FAILED",
  "details": [ {"field": "requestedTenureMonths", "code": "INVALID_TENURE"} ] }
```

5.6 Audit Record

Field	Content
requestId, timestamp, callerId	Who asked, when
inputHash and masked inputs	Tamper evidence without exposing raw PII
policyVersion and engineVersion	Exact rules and code used
outputs and reasonCodes	Complete numeric result

Field	Content
aiPromptVersion, aiModel, aiOutputHash, verifierResult	Explanation lineage and verification result
humanOverride (who, why)	Any manual change to the automated decision

5.7 Versioning and Backward Compatibility

Rule [Proposed]	Detail
Semantic API versions	Path-based (/v1, /v2); breaking changes only in a new major version
Additive changes	New optional input or output fields are allowed within a version; clients must ignore unknown output fields
Strict inputs	Unknown input fields are rejected so typos do not silently change decisions
Policy versioning	Independent of API version; every output carries policyVersion and engineVersion
Deprecation	Announce, run both versions in parallel, then retire after an agreed window
Schema registry	Single source of truth for TypeScript types, JSON schemas and documentation

Replay depends on this discipline. A decision made under policy 2026.10.1 and engine 1.4.2 must be reproducible exactly years later, so old versions are archived with their test vectors.

6. Engineering Milestones and Implementation Roadmap

The specification lists six milestones [Spec]. The table expands each into deliverables, exit criteria and risks. Durations in the timeline are planning estimates [Proposed].

Figure 6.1: Indicative timeline (weeks)

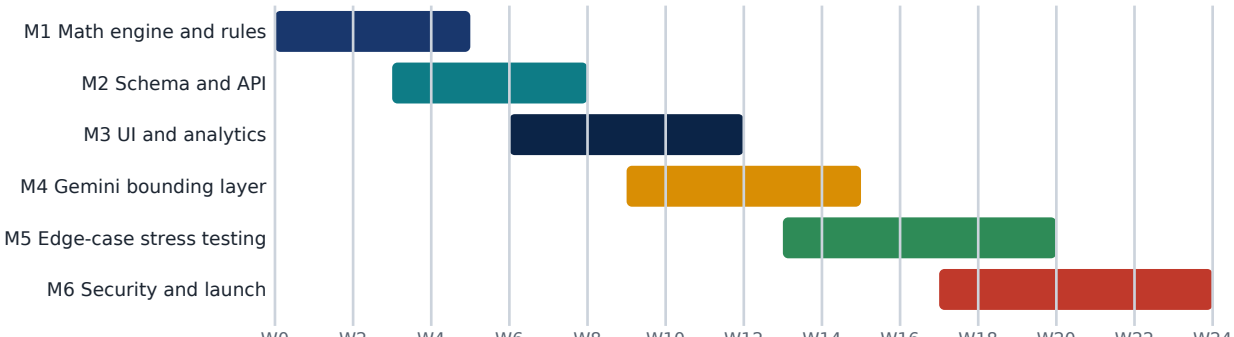


Figure 6.1: Milestones overlap; M5 and M6 run partly in parallel with M4 completion

6.1 Milestone 1: Mathematical Engine and Domain Rules

- **Spec:** defined credit equations, threshold lookup tables, built the core calculation library.
- **Deliverables:** annuity functions, FOIR and DTI calculators, policy table loader, rounding policy, reason codes.
- **Exit criteria:** all four scenarios reproduce spec outputs; property tests pass; zero-rate branch tested.
- **Risks:** ambiguous policy (missing SCN-03 and SCN-04 inputs); resolve with credit policy owners before coding.

6.2 Milestone 2: Schema Validation and API Integration

- **Spec:** established data contracts and input sanitization routines.

- **Deliverables:** strict schemas, sanity rules, API gateway integration, idempotency, structured errors.
- **Exit criteria:** fuzz tests show no malformed payload reaches the engine; contract tests pass for each client.
- **Risks:** unit confusion (lakh versus rupees), silently ignored unknown fields.

6.3 Milestone 3: Underwriting UI and Analytics Dashboard

- **Spec:** designed interactive financial gauges, debt ratio charts and risk status indicators.
- **Deliverables:** FOIR gauge, composition chart, scenario slider, status badges, accessibility pass.
- **Exit criteria:** charts match engine output exactly; usability review with credit officers.
- **Risks:** client-side recomputation causing mismatches; unclear visual encoding of risk.

6.4 Milestone 4: Gemini AI Bounding Layer

- **Spec:** connected calculation outputs to structured AI prompt templates for hallucination-free explanations.
- **Deliverables:** prompt templates, response schema, numeric verifier, template fallback, timeout handling.
- **Exit criteria:** 100% numeric-match on the evaluation set; fallback tested; injection tests pass.
- **Risks:** the 'hallucination-free' goal is not literally guaranteed; define it as verified numeric faithfulness plus human sampling.

6.5 Milestone 5: Edge-Case Stress Testing

The specification mentions evaluation over 500+ synthetic edge-case profiles. A suggested composition for that set:

Category	Count (min) [Proposed]	Examples
Boundary values	100	Income at minimum, FOIR exactly at cap, DTI exactly 40%
Extreme ratios	80	Obligations above income, huge requests, zero obligations
Rate and tenure extremes	80	6-month and 360-month tenures, rate near 0 and near 60
Variable income	80	Zero-income months, spikes, short history
Rounding traps	60	Ratios at 39.99 to 40.01, EMI half-rupee cases
Adversarial inputs	60	Negative numbers, NaN, strings, injection text, huge numbers
Regression goldens	40+	Four scenarios plus past production cases

- **Metrics:** exact-match rate to golden outputs (target 100%), p95 and p99 latency, error-class coverage.
- **Load test:** sustained request rate with the AI stage degraded, to confirm that the numeric path stays within budget.

6.6 Milestone 6: Security Hardening and Production Launch

- **Spec:** finalized audit logging, PII dynamic masking and enterprise deployment.

- **Deliverables:** immutable audit store, masking layer, key management, penetration test, runbooks, monitoring, disaster recovery.
- **Exit criteria:** penetration test findings closed; recovery drill completed; compliance sign-off (Section 7).
- **Risks:** late compliance findings; mitigate by starting reviews at Milestone 2.

6.7 Definition of Done (All Milestones)

Gate	Requirement
Quality	Unit, property and golden tests green; coverage target agreed
Security	Static analysis and dependency scan clean; secrets managed
Documentation	Contract, policy version and runbook updated
Review	Credit policy owner and risk/compliance sign-off
Operations	Dashboards and alerts live

6.8 Roles and Responsibilities (RACI)

Activity	Engineering	Credit policy	Risk / compliance	Product
Formulas and engine code	R	C	C	I
FOIR, DTI and stability tables	C	A / R	C	I
Schema and API contract	A / R	I	C	C
AI prompts and verifier	A / R	C	C	I
Regulatory mapping	I	C	A / R	C
Test set and golden outputs	R	C	A	I
Launch approval	R	C	A	R

R = responsible, A = accountable, C = consulted, I = informed. The key point is that policy numbers belong to credit policy and risk, not to engineers; engineers implement and test them.

7. Security and Compliance Standards

The specification states that decisions combine Deterministic Engine Safety with AI Contextual Guidance and that financial decisions are grounded in verified mathematics, ensuring audit readiness [Spec]. This section turns that statement into concrete controls and maps them to the Indian regulatory context.

Figure 7.1: Layered control model



Figure 7.1: Defense in depth across four layers

7.1 Threat Model

Threat	Impact	Control
Tampered inputs (inflated income)	Wrong approvals	Document verification upstream; anomaly flags; sanity rules
Prompt injection via applicant text	AI says something false or harmful	Only validated numbers in prompts; no free text; schema output; verifier
Policy table tampering	Systematic mispricing of risk	Signed configuration; dual approval; version in each record
Data leakage	Privacy breach and penalties	Encryption, masking, least privilege, access logging
Replay or duplicate submissions	Double processing	Idempotency keys; request IDs
Insider misuse	Unauthorized overrides	RBAC, separation of duties, immutable audit log
Model drift or provider change	Changed explanation quality	Pinned model version; regression prompts; fallback templates

7.2 Data Protection and PII Masking

- **Encryption:** TLS for all transport; strong encryption at rest; keys in a managed key service with rotation.
- **Dynamic masking:** show only the minimum necessary fields by role (for example, officers see income; support staff see masked identifiers).
- **Minimization:** the AI layer receives numbers and reason codes only, never names, contact details or identity numbers.
- **Retention:** defined retention per record type, with deletion workflows and legal-hold support.
- **Biometrics:** not stored by the platform; recent RBI digital-lending rules also prohibit storing biometric data for these purposes.

7.3 Access Control and Audit Logging

Role	Permissions [Proposed]
Credit officer	View decisions and notes; request replay; no policy edits
Risk analyst	View aggregate analytics; propose policy changes
Policy approver	Approve and sign policy versions (separate from proposer)
Engineer / operator	No access to raw PII in production; break-glass with approval and logging
Auditor	Read-only access to audit records and replay

Audit records (Section 5.6) are written to an append-only store with hash chaining, so any later alteration is detectable. Every manual override records who changed what and why.

7.4 Regulatory Context: RBI Digital Lending Directions, 2025

Credify AI targets Indian lenders, so the main regulatory reference is the Reserve Bank of India (Digital Lending) Directions, 2025. The following points come from RBI's circular and law-firm summaries of it; the sources are listed at the end of this section.

Topic	What the Directions say (summary)	Credify AI implication
Date and scope	Issued 8 May 2025; apply to regulated entities such as banks, NBFCs and housing finance companies across the lending lifecycle, including credit assessment and approval	The platform supports regulated entities and is within the covered process
Key Facts Statement (KFS)	Borrowers must receive a clear Key Facts Statement before the loan agreement is executed, including rate, fees and penal charges	Decision output must carry data for the KFS: EMI, tenure, total cost

Topic	What the Directions say (summary)	Credify AI implication
Direct flows	Loan disbursal and repayment must be directly between borrower and the regulated entity	No fund-flow logic inside Credify; decision engine only
Outsourcing responsibility	Outsourcing to service providers does not remove the regulated entity's responsibility for outcomes	Lender stays accountable; provide transparency and audit access
Data protection	Strong data protection and cybersecurity expected; biometric data storage prohibited	Controls in Sections 7.2 and 7.3
Digital lending apps	Apps must be reported to RBI on its portal (deadline 15 June 2025)	Lender-side duty; document which apps use Credify
Credit reporting	Digital lending, including short-term and deferred-payment credit, must be reported to credit information companies	Provide structured outputs for bureau reporting
Multi-lender arrangements	New requirements effective 1 November 2025	Check applicability if offers span lenders
Default loss guarantees	Must not replace strong credit underwriting; revolving credit and credit-card DLGs not allowed	Reinforces the need for a real capacity engine

Verification required. Regulations change and this summary is not legal advice. Before launch, the lender's compliance and legal teams should confirm the current text of the Directions and any later amendments, together with data-protection law (for example the Digital Personal Data Protection Act, 2023 and its rules) and the lender's own fair-practices code.

Sources used: Reserve Bank of India circular index entry for the Digital Lending Directions, 2025; summaries by Kochhar and Co. (Mondaq), IndiaLaw LLP, Legal500 and SCC Online.

7.5 AI Governance and Fairness

- **Decision authority:** the AI layer has none. This is documented and tested, which simplifies model-risk review.
- **Explainability:** every decision carries reason codes that map to plain-language explanations.
- **Fairness monitoring:** track approval and conditional rates by permitted segments; investigate unexplained gaps in policy outcomes.
- **Protected attributes:** not inputs to the engine or to the prompts.
- **Human oversight:** declines and conditional cases can be reviewed by an officer; overrides are logged.
- **Grievance redress:** applicants can contest a decision; replay shows the exact inputs and rules used.

7.6 Claims Review: 'Deterministic 100% Exact + AI 100% Transparent'

The specification's tagline claims 100% exactness and 100% transparency. A defensible, verifiable restatement is:

Original claim	Defensible restatement [Proposed]
100% exact calculation	All figures are produced by deterministic, versioned code and are reproducible from stored inputs
Hallucination-free explanations	AI text is generated from verified inputs and is automatically checked so that every figure matches the engine; mismatches fall back to templates
100% transparent AI guidance	Every decision lists reason codes, the policy version and the AI prompt and model version used
Full compliance	Controls are designed to support compliance; final compliance is confirmed by the lender's legal review

7.7 Operational Resilience

- **Availability:** stateless engine instances behind a load balancer; multi-zone deployment.
- **Monitoring:** latency percentiles, error rates, verifier mismatch rate, fallback rate, policy-version distribution.
- **Incident response:** runbooks for engine errors, AI outages and suspected data exposure.
- **Disaster recovery:** backups of audit store and policy table; periodic restore drills.

8. Conclusion, Risks and Future Work

8.1 Conclusion

The Credify AI architecture establishes a fast, reliable and transparent standard for modern credit underwriting [Spec]. By pairing deterministic calculations with generative AI guidance, it can deliver sub-second automated loan evaluations for financial institutions. The expansion in this document makes the design more precise in three ways: it fills policy gaps (FOIR tiers, DTI bands, stability factor), it supplies fully worked and test-verified scenarios, and it translates broad claims into controls that can be tested and audited.

8.2 Key Findings from the Review

- The mathematics in the specification are correct where inputs are complete (SCN-01, SCN-02) and reproducible once missing assumptions are supplied (SCN-03, SCN-04).
- The product cap of INR 50,00,000 can bind before applicant capacity does (SCN-01), which should be explained to applicants.
- Mitigation can be computed automatically: extending SCN-03 to 300 months moves it from CONDITIONAL to APPROVED.
- Tagline claims should be restated in verifiable terms (Section 7.6).
- The status union should be reconciled with the CAP ADJUSTED outcome (Section 5.3).

8.3 Risk Register

Risk	Likelihood	Impact	Mitigation
Ambiguous policy inputs	High	High	Policy owner sign-off on FOIR, DTI bands and stability rules before build
Rate data staleness	Medium	High	Timestamp rates; refresh rules; sensitivity display
AI numeric mismatch	Medium	Medium	Verifier and template fallback; monitor mismatch rate
Regulatory change	Medium	High	Compliance watch; configurable disclosures
Latency from AI stage	High	Medium	Asynchronous advisory; timeouts
Income misstatement	Medium	High	Upstream verification; anomaly flags
Bias in outcomes	Low-Medium	High	Fairness monitoring; excluded attributes; review

8.4 Future Work

- Cash-flow underwriting from bank-statement data with consent-based data sharing.
- Product-specific rate engines and risk-based pricing.
- Portfolio-level analytics: stress tests of approved loans against rate and income shocks.
- Multilingual explanations with identical verified numbers.

- Counter-offer generator that proposes the best amount and tenure combination automatically.
- Challenger models (statistical scorecards) run in shadow mode to compare with rules.

8.5 Success Metrics

Metric	Target [Proposed]
Numeric decision latency (p95)	under 850 ms end to end; engine under 10 ms
Golden-test exact match	100%
AI numeric-match rate	100% after verifier (raw model rate tracked)
Decisions with reason codes	100%
Manual-review rate	tracked, trending down
Applicant clarity score (survey)	above agreed baseline

Appendix A: Formula Sheet

Quantity	Formula
Monthly rate	r = annual rate / 12 / 100
EMI	E = P x r / (1 - (1 + r)^-n)
Max principal	P = E x (1 - (1 + r)^-n) / r
Adjusted income	I_adj = I x stability factor
FOIR cap (rupees)	Cap = I_adj x FOIR percentage
Available EMI	A = Cap - existing obligations
Post-loan DTI	(existing + requested EMI) / gross income x 100
Total interest	E x n - P
Zero-rate case	E = P / n; P = E x n
Evaluated principal	min(max principal, 50,00,000)

Appendix B: Glossary

Term	Meaning
EMI	Equated Monthly Installment: fixed monthly repayment of a loan
FOIR	Fixed Obligation to Income Ratio: share of income allowed for all fixed debt payments
DTI	Debt-to-Income ratio: monthly obligations divided by gross monthly income
Principal	Amount borrowed, excluding interest
Tenure	Repayment period, here in months
Stability Factor	Multiplier below 1 that discounts variable income
KFS	Key Facts Statement: standard loan disclosure given before agreement
DLG	Default Loss Guarantee: a third-party guarantee against loan defaults
RE	Regulated Entity: bank, NBFC or similar lender under RBI rules
LSP / DLA	Lending Service Provider / Digital Lending App
PII	Personally Identifiable Information
Fail closed	On error, deny or defer rather than allow

Appendix C: Specification Review Checklist

Item	Status	Owner action
State FOIR for SCN-04 (50%)	Open	Policy owner
Provide rate, tenure, existing debt for SCN-03	Open	Product team
Define DTI comfort band (40%) and its effect	Open	Credit policy
Reconcile status values with CAP ADJUSTED	Open	Engineering
Define foirPercentage versus DTI naming	Open	Engineering
Restate tagline claims verifiably	Open	Product and compliance
Confirm AI model name and version policy	Open	AI lead
Confirm regulatory text with legal counsel	Open	Compliance

Appendix D: Extended Applicant (Candidate) Research

This appendix deepens the archetypes of Section 1.7. For each, it lists what to verify, typical failure patterns, how the engine treats the file and what the advisory layer should say. Policy values are proposals [Proposed].

D.1 Salaried Applicant (SCN-01, SCN-03)

Dimension	Detail
Income evidence	Recent payslips, bank credits, employer verification; use fixed gross, treat bonuses separately
Main risks	Over-leverage through many small loans; recent job change; income concentrated in variable pay
Engine treatment	FOIR 55% (SCN-01); DTI comfort band 40%; score floor applied
Typical mitigations	Longer tenure, smaller amount, co-applicant, closing small loans
Advisory message	Explain headroom, show how tenure changes EMI and total interest

D.2 Self-Employed or Retailer (SCN-02)

Dimension	Detail
Income evidence	Tax returns, GST filings, bank statements; averaged over 12 to 24 months
Main risks	Mixing personal and business debt; seasonal cash flow; informal obligations not captured
Engine treatment	FOIR 50%; obligations must include business loans serviced from personal income
Typical mitigations	Consolidate loans, reduce obligations before reapplying, secured product
Advisory message	Give a concrete debt-reduction target (about INR 15,622 for the SCN-02 example)

D.3 Gig Worker or Consultant (SCN-04)

Dimension	Detail
Income evidence	12 months of platform payouts or invoices; consistency matters more than peak months
Main risks	Income volatility; dependence on one platform or client; no employer verification
Engine treatment	Stability Factor (0.75 in spec); FOIR 50% [Assumed]; capacity shown on adjusted income
Typical mitigations	Longer history, second income source, shorter tenure at lower amount, co-applicant
Advisory message	Explain that reported income was discounted for variability and how to improve the factor

D.4 Thin-File or Young Applicant

Dimension	Detail
Income evidence	Offer letter, early payslips, education or training records where policy allows
Main risks	Little repayment history; short employment; unstable address and job
Engine treatment	Conservative FOIR (45%) and lower amount ceiling [Proposed]; weight capacity over score
Typical mitigations	Small starter loan, guarantor or co-applicant, secured credit
Advisory message	Describe a credit-building roadmap with timing, without promising approval

D.5 Cross-Archetype Comparison

Archetype	FOIR	Income adjustment	Verification effort	Typical status
Salaried	55%	None	Low	Approved or conditional

Archetype	FOIR	Income adjustment	Verification effort	Typical status
Self-employed	50%	Averaging	High	Declined or conditional when leveraged
Gig / consultant	50%	Stability factor	High	Cap adjusted
Thin-file	45%	None, lower ceiling	Medium	Conditional with co-applicant

D.6 Co-Applicant Effect (Illustration)

Adding a co-applicant raises combined income and combined obligations. For SCN-03 at 240 months and 41.2% DTI, a co-applicant with income INR 40,000 and obligations INR 5,000 gives combined income 1,50,000 and obligations 22,520 plus the new EMI 27,769: DTI = 50,289 / 1,50,000 = 33.5%, which falls in the comfortable band. The engine should compute this directly when a co-applicant is supplied.

Appendix E: Additional Edge-Case Test Vectors

The following vectors were executed against the reference engine of Section 3.10. They illustrate the style of the 500+ synthetic profiles of Milestone 5.

ID	Case	Inputs (income, debt, type, rate, months, request)	Result
T1	FOIR boundary, no request	1,00,000; 30,000; salaried; 10%; 120; none	Available EMI 25,000; max loan 18,91,779; APPROVED
T2	Very low income	15,000; 0; salaried; 12%; 60; 2,00,000	Available EMI 8,250; EMI 4,449; DTI 29.7%; APPROVED
T3	Request equals cap	4,00,000; 0; salaried; 9%; 240; 50,00,000	Max loan 50,00,000 (cap); EMI 44,987; DTI 11.2%; APPROVED
T4	Stretched DTI	1,00,000; 10,000; salaried; 10%; 120; 30,00,000	EMI 39,646; DTI 49.6%; CONDITIONAL (within FOIR, above band)
T5	Gig, no debt	60,000; 0; gig; 14%; 48; stability 0.6	Capacity 18,000; max loan 6,58,701; CAP ADJUSTED
T6	Zero interest rate	1,00,000; 0; salaried; 0%; 60; 20,00,000	EMI 33,334 (rounded up); DTI 33.3%; APPROVED

Property checks to automate: (1) repaying the computed EMI each month leaves a balance of at most one rupee after the last installment; (2) max_principal applied to emi(P) returns P within rounding; (3) raising income never reduces available EMI; (4) raising the rate never increases max principal; (5) when requested EMI is at most available EMI and DTI is at most 40%, status is APPROVED.

Appendix F: Sample Advisory Outputs

The AI layer writes only from engine results. These samples show target tone and structure. All figures are copied from the engine output and would be checked by the verifier.

F.1 SCN-03 (CONDITIONAL)

```
Summary: Your requested loan fits within your repayment capacity, but it
would use 41.2% of your monthly income, which is above our comfort level.
Risk notes:
 - Your monthly installment would be about INR 27,769.
 - You can afford up to about INR 42,980 per month in installments.
Ways to move forward:
 - Choose a 25-year term: the installment falls to about INR 26,003.
 - Reduce the amount to about INR 28.6 lakh.
 - Apply with a co-applicant whose income can be added.
```

F.2 SCN-02 (DECLINED)

```
Summary: At this time your existing monthly payments are already higher
than the share of income we can allow, so we are unable to offer this loan.
Risk notes:
 - Your existing payments are INR 42,000; the allowed limit is INR 37,500.
Ways to move forward:
 - Reduce monthly payments by at least INR 4,500 to reach the limit, and
   by about INR 15,622 to have room for a INR 5,00,000 loan.
 - Consider consolidating loans or adding a co-applicant.
```

Guardrails. Samples avoid promises (no 'you will be approved'), avoid personal attributes and state one concrete next step per bullet.

Appendix G: Deployment Architecture and Operations

Figure G.1: Deployment view

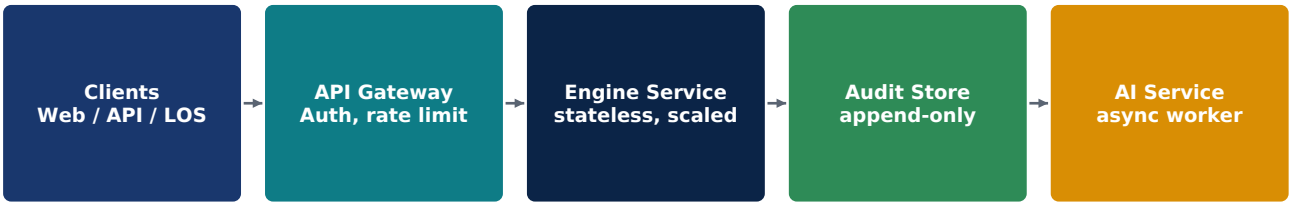


Figure G.1: The numeric path (gateway, engine, audit) is separate from the AI worker

Component	Scaling and resilience
API gateway	Horizontal; authentication, throttling, request IDs
Engine service	Stateless; autoscale on CPU and latency; multi-zone
Policy store	Signed, versioned; cached with last-known-good fallback
Audit store	Append-only, replicated, hash-chained; restore drills
AI worker	Queue-based; timeouts; circuit breaker to template fallback
Observability	Metrics, traces and alerts on latency, mismatch rate, fallback rate

G.1 Release and Change Management

- Policy changes follow propose, review, approve, sign, schedule, then activate on an effective date.
- Engine releases run the golden suite and a shadow comparison against the previous version before promotion.
- Rollback restores the previous policy and engine version; past decisions remain replayable under their own versions.

Appendix H: Amortization Illustration

INR 50,00,000 at 10.5% over 240 months; EMI 49,919 (rounded up). First 12 months:

Month	EMI	Interest	Principal	Balance
1	49,919	43,750	6,169	49,93,831
2	49,919	43,696	6,223	49,87,608
3	49,919	43,642	6,277	49,81,331
4	49,919	43,587	6,332	49,74,998
5	49,919	43,531	6,388	49,68,610
6	49,919	43,475	6,444	49,62,167

Month	EMI	Interest	Principal	Balance
7	49,919	43,419	6,500	49,55,667
8	49,919	43,362	6,557	49,49,110
9	49,919	43,305	6,614	49,42,496
10	49,919	43,247	6,672	49,35,823
11	49,919	43,188	6,731	49,29,093
12	49,919	43,130	6,789	49,22,303

In the first year the borrower pays about INR 5,99,028 in total, of which about INR 5,21,332 is interest and only about INR 77,696 reduces principal. This is why the Key Facts Statement and the advisory notes must show total cost and not only the monthly installment.